A

Report on

PRACTICAL SECURITY ASSESSMENT (TASK 7)

Submitted In Partial Fulfillment of the Requirements

for the

Cyber Security Internship (Week 2) (VAPT)

Internship Organization

CyArt Tech LLP

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LIST OF FIGURES

CHAPTER NO.	TITLE	PAGE NO.
1	Figure 1 : Vapt flow diagram	2
2	Figure 2: Network scanning using nmap	2
3	Figure 3: Authenticated vs Unauthenticated Scanning	3
4	Figure 4: Measure the severity of vulnerabilities	4
5	Figure 5: CVSS Vulnerability Scoring System	5
6	Figure 6 : Penetration Testing Lifecycle	5
7	Figure 7: Reconnaissance using Shodan	6
8	Figure 8: Exploitation using Metasploit Framework	6
9	Figure 9: Web Application Scanning using nikto	8
10	Figure 10: SQL Injection Attack on DVWA	9
11	Figure 11: Cross-Site Scripting (XSS) Attack in DVWA	9
12	Figure 12: Mitigation Techniques	11
13	Figure 13: Exploit Development Process	11

TABLE OF CONTENTS

CHAPTER NO.	TITLE	PAGE NO.
	LIST OF FIGURES	**i**
	TABLE OF CONTENTS	**ii**
1.	**INTRODUCTION**	
	1.1 OVERVIEW	1
	1.2 PURPOSE OF THIS TASK	1
	1.3 TARGET INFORMATION	1
	1.4 OBJECTIVES	2
2.	**VULNERABILITY SCANNING TECHNIQUES**	
	2.1 OVERVIEW	2
	2.2 TYPES OF SCANNING	3
	2.3 VULNERABILITY SCORING (CVSS V4.0)	4
	2.4 FALSE POSITIVES	4
	2.5 KEY OBJECTIVES	4-6
3.	**PENETRATION TESTING TECHNIQUES**	
	3.1 OVERVIEW	7
	3.2 PHASES OF PENETRATION TESTING	7
	3.3 METHODOLOGIES	7
	3.4 ETHICAL CONSIDERATIONS	7-8
	3.5 KEY OBJECTIVES	8-9
4.	**AUTOMATION AND NETWORK DIAGRAM**	
	4.1 OVERVIEW	10
	4.2 TYPES OF EXPLOITS	10
	4.3 EXPLOIT DEVELOPMENT PROCESS	10
	4.4 MITIGATION TECHNIQUES	11
	4.5 KEY OBJECTIVES	11
5.	**CONCLUSION AND REFERENCE**	
	5.1 CONCLUSION	12
	5.2 REFERENCE	12

1. INTRODUCTION

1.1 Overview

Vulnerability Assessment and Penetration Testing (VAPT) is a comprehensive cybersecurity process used to identify, analyze, and mitigate vulnerabilities in systems, networks, and applications. It combines automated scanning tools with manual testing techniques to simulate real-world cyberattacks.

In today's digital environment, organizations rely heavily on IT infrastructure, making them vulnerable to cyber threats. VAPT helps in detecting security weaknesses before attackers exploit them, thereby strengthening the overall security posture.

1.2 Purpose of This Task

The main purpose of this task is to perform a security assessment of a target system using VAPT techniques. This includes identifying vulnerabilities, analyzing risks, and suggesting appropriate mitigation strategies.

This task also aims to provide practical exposure to cybersecurity tools and methodologies used in real-world scenarios.

1.3 Target Information

The target system used in this assessment is a vulnerable virtual machine configured in a controlled lab environment.

Target System: Metasploitable / Test Machine

Attacker Machine: Kali Linux

Network Type: Virtual Network (NAT/Host-only)

Tools Used: Nmap, Nikto, Metasploit

1.4 Objectives

- Identify vulnerabilities in the system
- Analyze security risks
- Perform penetration testing
- Suggest mitigation techniques
- Improve overall system security

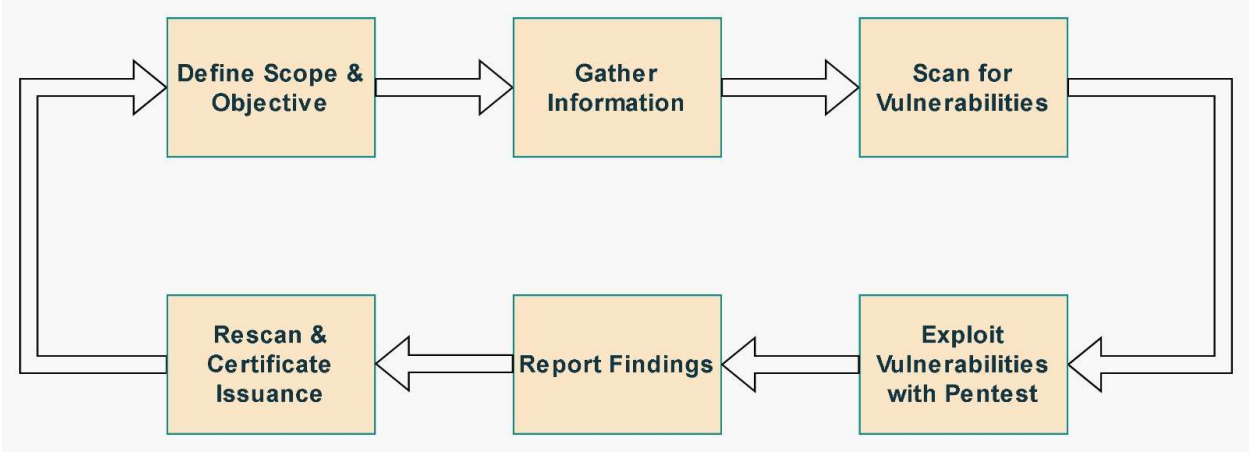

Figure 1: Vapt flow diagram

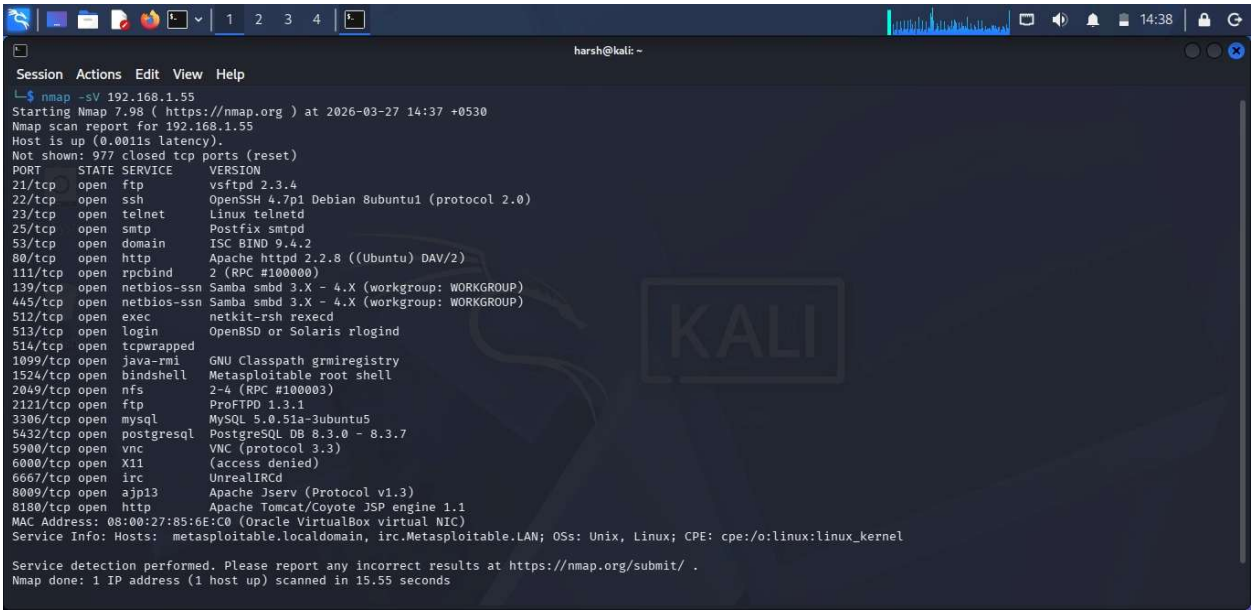

Figure 2: Network scanning using nmap

2. VULNERABILITY SCANNING TECHNIQUES

2.1 Overview

Vulnerability scanning is an automated process used to detect known security weaknesses in a system. It helps in early identification of vulnerabilities and reduces the risk of exploitation.

2.2 Types of Scanning

1. Network Scanning

- Network scanning involves identifying active devices, open ports, and running services within a network.
- Tool Used: Nmap
- Example: Detecting open ports such as
- Port 21 (FTP)
- Port 22 (SSH)
- Port 80 (HTTP)

2. Application Scanning

- Application scanning focuses on detecting vulnerabilities in web applications.
- Tool Used: Nikto
- It detects:
- Outdated software
- Server misconfigurations
- Security vulnerabilities

Type	Description
Authenticated	Uses login credentials for deep scanning
Unauthenticated	Simulates an external attacker

Figure 3: Authenticated vs Unauthenticated Scanning

2.3 Vulnerability Scoring (CVSS v4.0)

The Common Vulnerability Scoring System (CVSS) is used to measure the severity of vulnerabilities.

Score Range	Severity Level
0.0 – 3.9	Low
4.0 – 6.9	Medium
7.0 – 8.9	High
9.0 – 10.0	Critical

Figure 4: Measure the severity of vulnerabilities

Example:
The WannaCry ransomware attack exploited a critical SMB vulnerability, causing widespread global damage.

2.4 False Positives

False positives occur when a vulnerability scanner incorrectly reports a vulnerability that does not actually exist. These must be manually verified to ensure accuracy.

2.5 Key Objectives

- Identify vulnerabilities
- Assess risk levels
- Prioritize fixes
- Improve security posture

CVSS RATING	
CVSS SCORE	QUALITATIVE RATING
0.0	None
0.1 - 3.9	Low
4.0 - 6.9	Medium
7.0 - 8.9	High
9.0 - 10.0	Critical

Figure 5: CVSS Vulnerability Scoring System

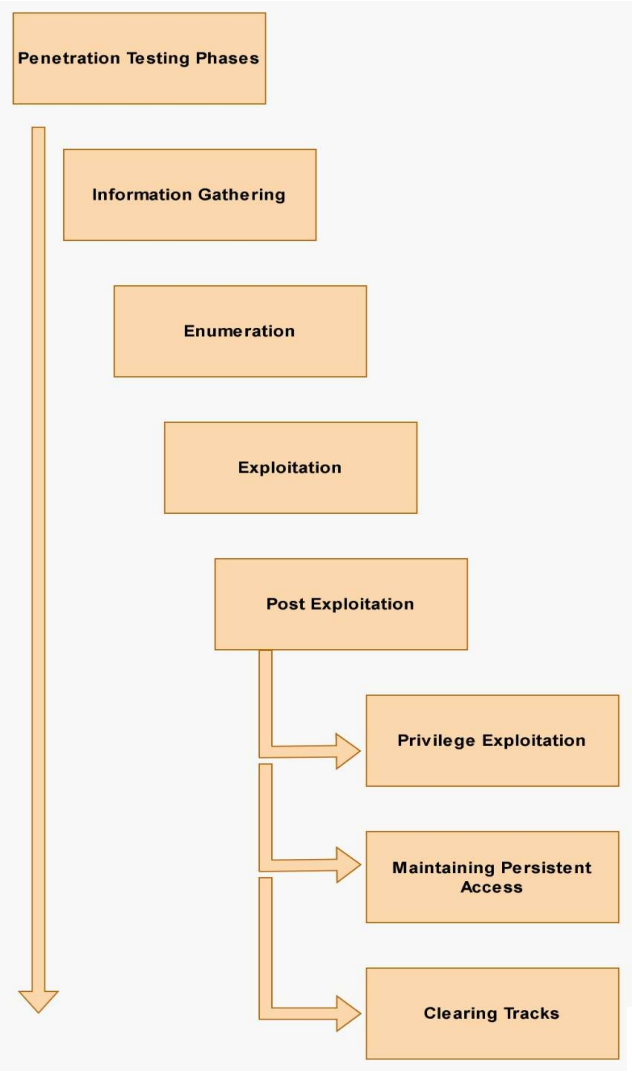

Figure 6: Penetration Testing Lifecycle

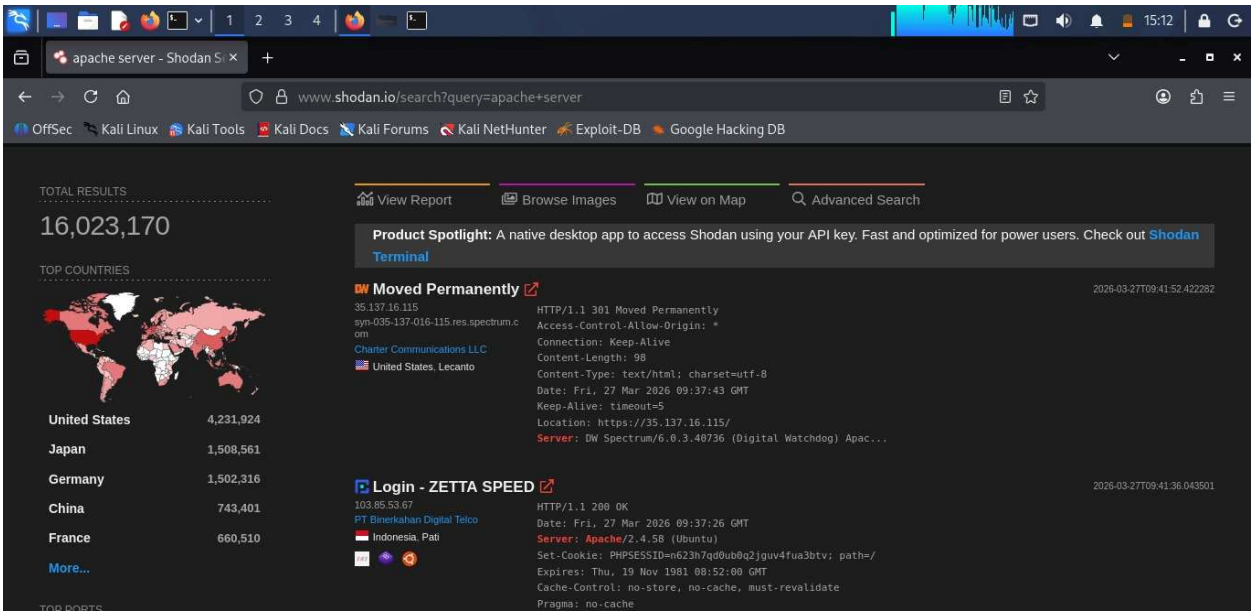

Figure 7: Reconnaissance using Shodan

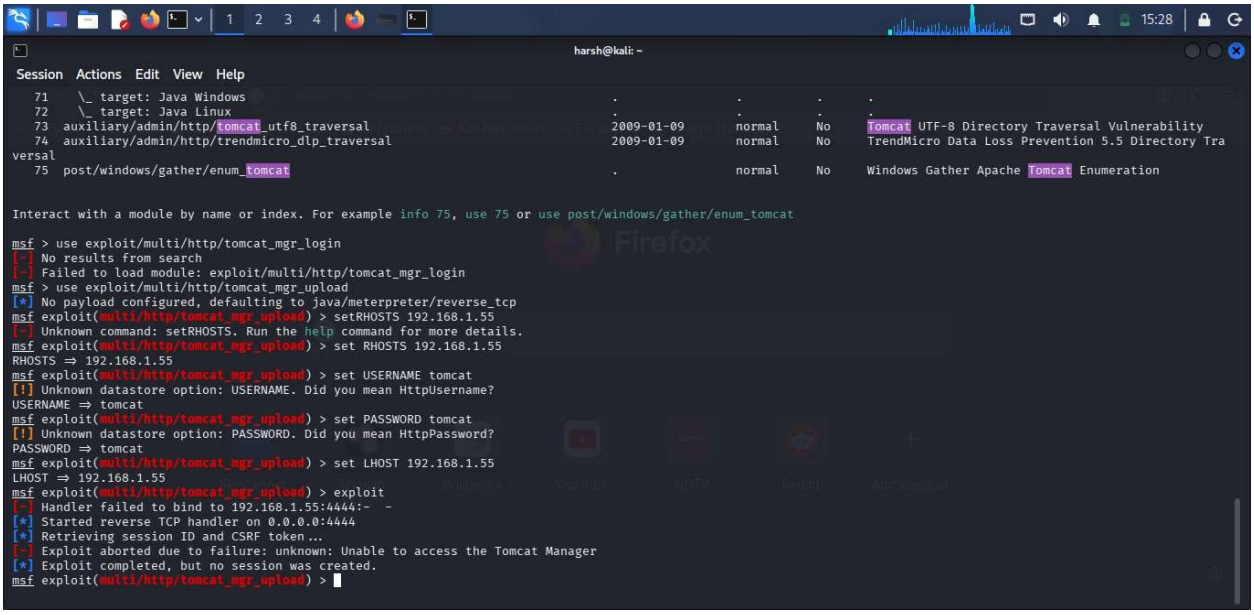

Figure 8: Exploitation using Metasploit Framework

3. PENETRATION TESTING TECHNIQUES

3.1 Overview

Penetration testing is a simulated cyberattack performed to evaluate the security of a system by exploiting identified vulnerabilities.

3.2 Phases of Penetration Testing

1. Reconnaissance

Information gathering about the target system.

Tool: Shodan

2. Scanning

Identifying vulnerabilities using automated tools.

Tools: Nmap, Nessus

3. Exploitation

Attempting to exploit vulnerabilities.

Tool: Metasploit

4. Post-Exploitation

Maintaining access and extracting sensitive information.

5. Reporting

Documenting findings, risks, and recommendations.

3.3 Methodologies

- PTES (Penetration Testing Execution Standard)
- OWASP WSTG (Web Security Testing Guide)

These provide structured guidelines for performing penetration testing.

3.4 Ethical Considerations

- Obtain proper authorization

- Follow defined scope
- Avoid system damage
- Maintain confidentiality

3.5 Key Objectives

- Simulate real-world attacks
- Identify critical vulnerabilities
- Strengthen system defenses

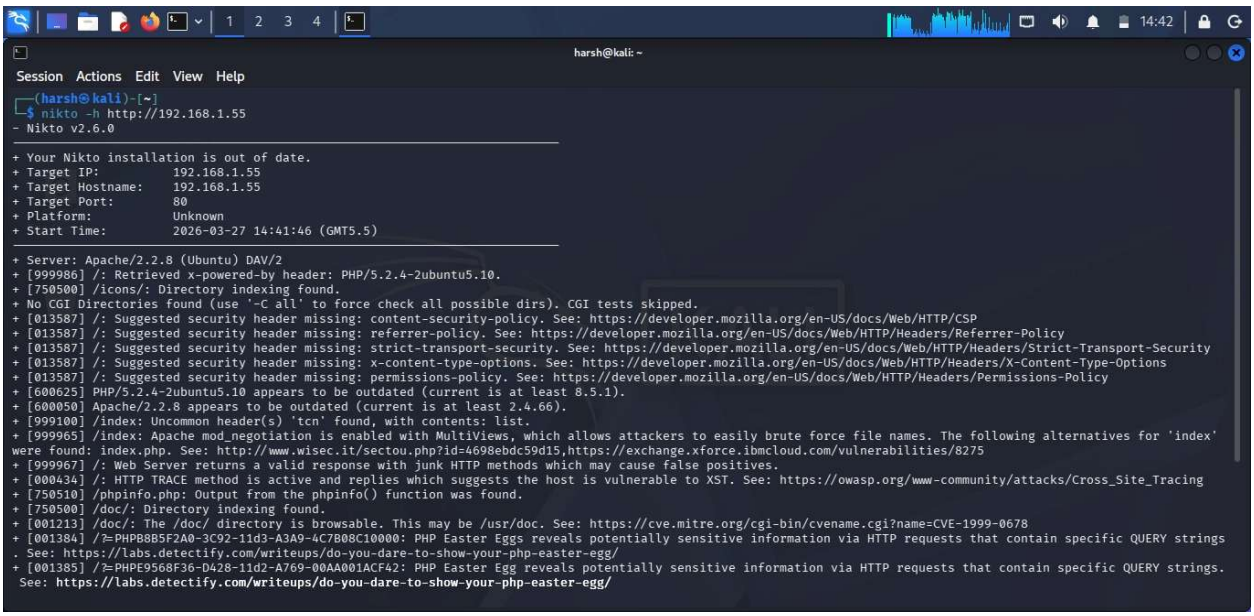

Figure 9: Web Application Scanning using nikto

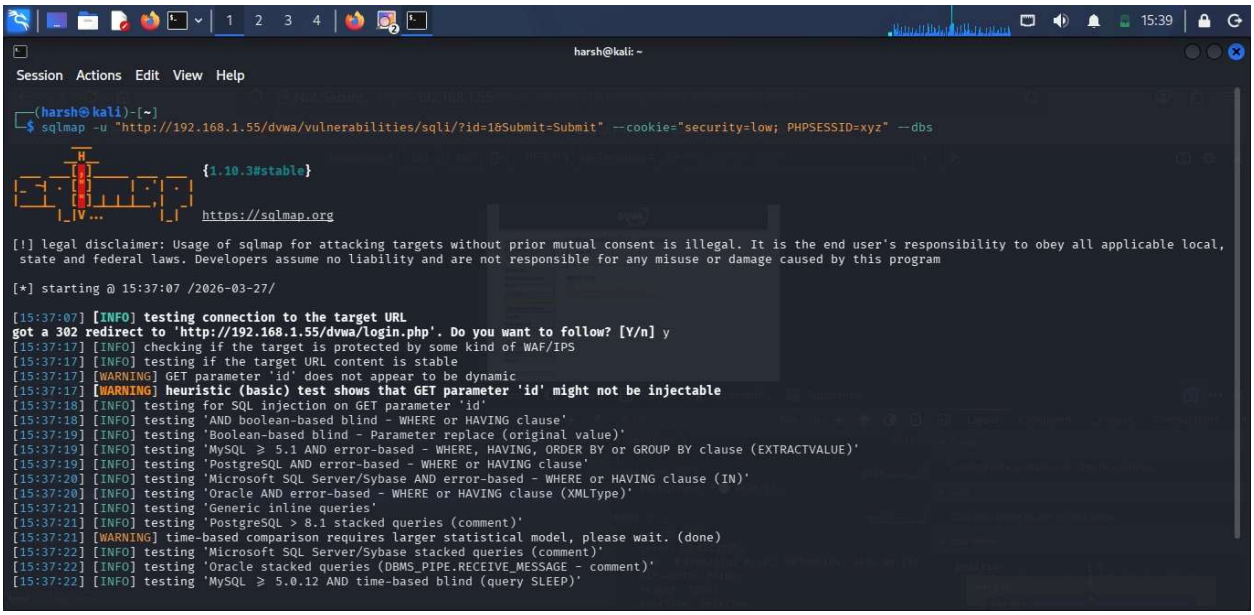

Figure 10: SQL Injection Attack on DVWA

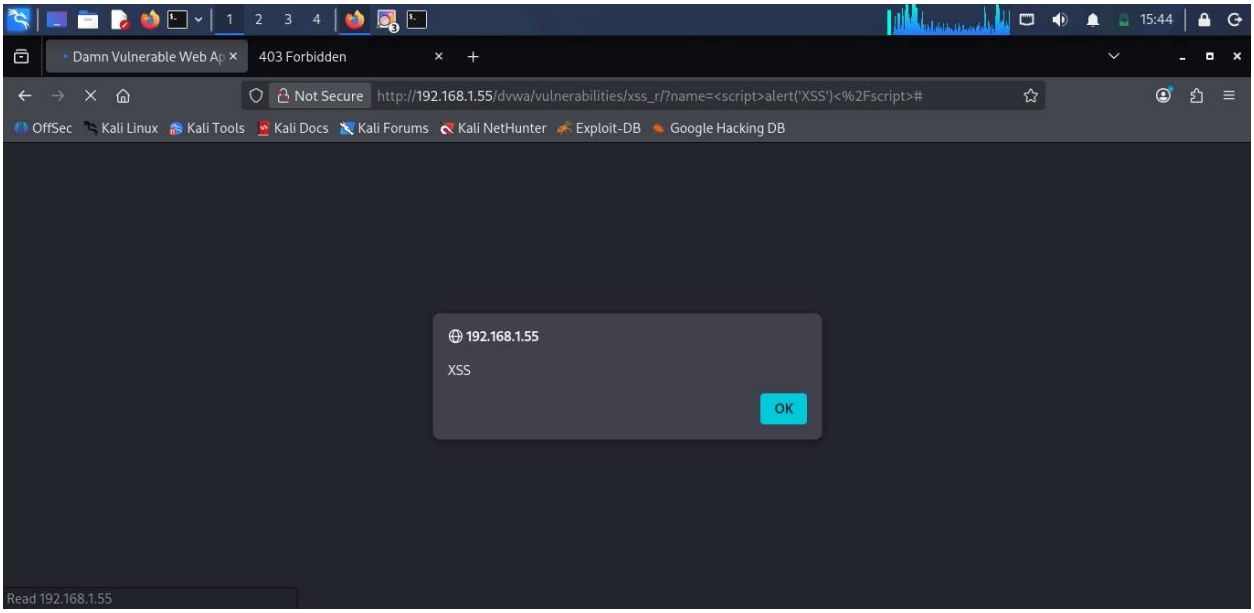

Figure 11: Cross-Site Scripting (XSS) Attack in DVWA

4. EXPLOIT DEVELOPMENT BASICS

4.1 Overview

Exploit development involves creating code that takes advantage of vulnerabilities to gain unauthorized access or perform malicious actions.

4.2 Types of Exploits

Buffer Overflow

Occurs when excess data overwrites memory, leading to system compromise.

SQL Injection

Allows attackers to manipulate database queries.

Example: ' OR 1=1 –

Cross-Site Scripting (XSS)

Allows injection of malicious scripts into web pages.

Example: <script>alert("Hacked")</script>

4.3 Exploit Development Process

- Identify vulnerability
- Analyze system behavior
- Develop Proof of Concept (PoC)
- Test in a controlled environment

4.4 Mitigation Techniques

Technique	Description
ASLR	Randomizes memory locations
WAF	Filters malicious traffic
Patching	Fixes vulnerabilities

Figure 12: Mitigation Techniques

4.5 Key Objectives

- Understand attack techniques
- Develop safe testing methods
- Improve application security

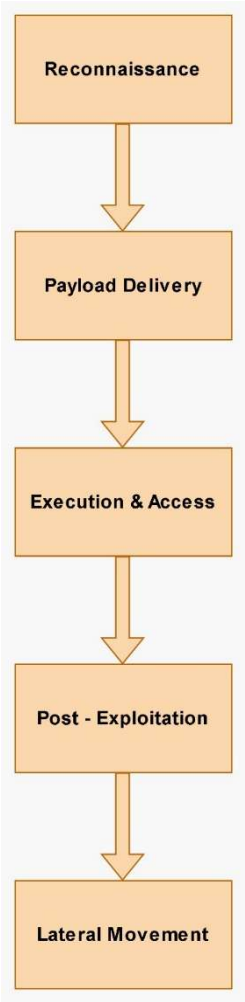

Figure 13: Exploit Development Process

5. CONCLUSION AND REFERENCE

5.1. Conclusion

Vulnerability Assessment and Penetration Testing (VAPT) is an essential cybersecurity practice that helps organizations identify and fix security weaknesses before attackers exploit them.

By combining automated scanning and manual testing, organizations can build a strong defense against cyber threats. Regular testing and continuous monitoring are necessary to maintain a secure environment.

5.2 References

OWASP Testing Guide

NIST SP 800-115

PTES Framework

Exploit-DB

TryHackMe